QM II: Homework 1

Lucas I. Klotz, GPS

ACKNOWLEDGMENTS

For this work, in addition to previous exposure to R programming as both a graduate Economics student in Brazil and a research assistant at Stanford University, and during QM II lectures, labs, and TA office hours, I, Lucas I. Klotz, acknowledge having worked with my classmate Mateus Zanini. I also acknowledge having used ChatGPT to troubleshoot R code. Specifically, ChatGPT assisted in adjusting the kernel density curve to use percentage frequencies rather than counts. Grammarly was used to correct and improve the writing. All interpretations, analyses, and conclusions in this assignment are my own, and I completed this work independently.

1 Preliminaries

Q1. - Q3.

dim() provides the number of observations and the number of variables of the dataset. In this case, there are 20,990 observations and 16 variables.

Q4.

No missing values present in the dataset.

2 Data and Descriptive Statistics

Q5. - Q6.

Following the Codebook, I would go with *wildfire2yr5000*, as it provides precisely what the prompt requests: an indicator of wildfire within the block group (BG) in the past 2 years. 66 BGs faced this problem.

The dependent variable (DV), *envbi*, has an average of approximately 0.63 and a variance of approximately 0.015. The independent variable (IV), *mindistkm* (*mindist5000* as km), has an average of approximately 60.4 km and a variance of approximately 2248.598 km.

The covariance between the two variables is approximately -0.897, which indicates a negative relationship between them.

3 Visualizing the Data

Q7.

Below I show the figures. Each figure's note describes my interpretation.

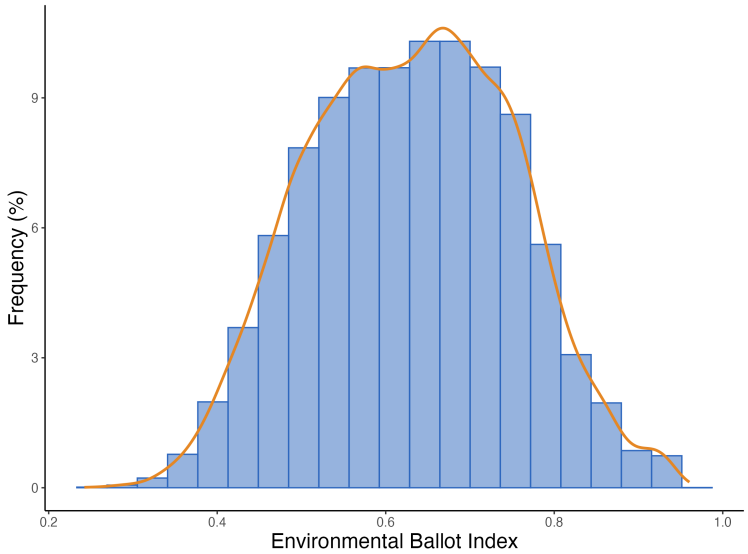

Figure 1: Histogram of Environmental Ballot Index

Notes: This figure shows the distribution of the Environmental Ballot Index across the Census Block Groups (BGs) in 2010 in California. The histogram shows that the scores are nearly normally distributed around the mean (0.63), with the majority of the voters supporting Proposition 23 in California's 2010 elections across the BGs. The kernel density curve smooths the distribution for easier interpretation. N = 20,990.

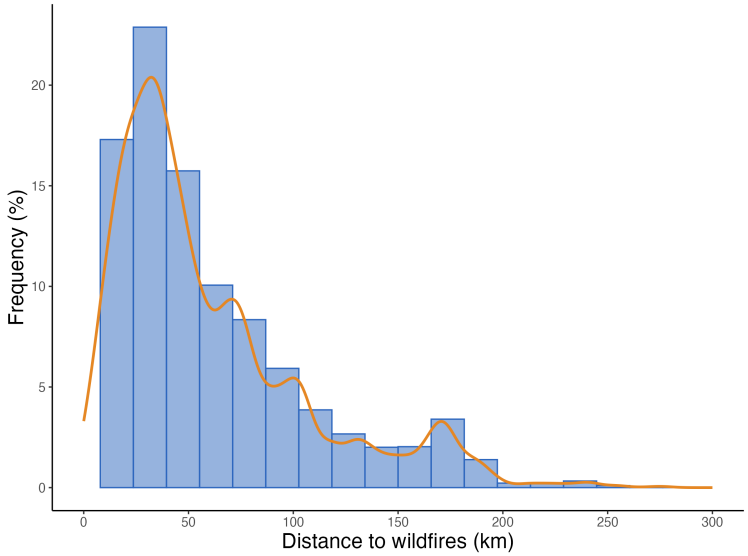

Figure 2: Histogram of Distance to Wildfires (km)

Notes: This figure shows the distribution of the closest distance to wildfires (in km) of significant size ($\geq$ 5,000 acres) in one or two years before 2010. The histogram shows a left-skewed distribution, indicating that the majority of wildfires were within 50 km of the BGs' ground zero, with a few extending to 100-200 km range. The kernel density curve smooths the distribution for easier interpretation. N = 20,990.

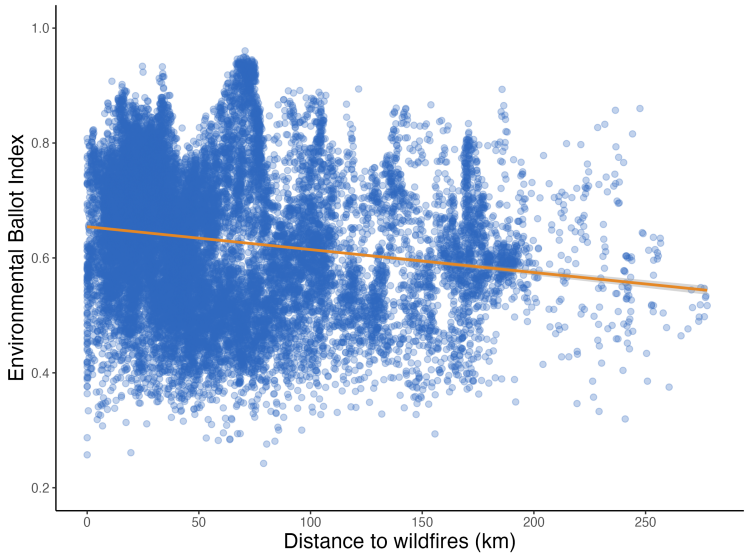

Figure 3: Environmental Ballot Index vs. Distance to Wildfires (km)

Notes: This figure shows the correlation between the Environmental Ballot Index and the distance to wildfires (in km) greater or equal to 5,000 acres across BGs in California. The scatterplot shows high variability in the Index score at each distance, but overall it shows a modest negative relationship (correlation = -0.15). In other words, as the distance to wildfires increases, the Index score falls. The regression line (in yellow) highlights this trend. N = 20,990.

Q8.

Table 1: Regression results

	Dependent variable:
	Envir. Index
Distance to wildfires (km)	-0.0004^{***}
	(0.00002)
Intercept	0.654^{***}
	(0.001)
Observations	20,990
R^2	0.025

Notes: This table shows the results from the bivariate regression estimation. The dependent variable is the Environmental Ballot Index. The independent variable is the BG's distance to wildfires (in km). N = 20,990. *p<0.1; **p<0.05; ***p<0.01.

The estimated relationship is approximately -0.0004. It is statistically significant at the 1% level, as the p-value is well below 0.01.

The R^2 is 0.025, which means that 2.5% of the variation on the Environment Index (DV) is explained by the variation of the distance to the wildfires (IV). Therefore, the current model specification does not do a very good job at explaining the variation of the DV.

Q9.

They are identical. $lm()$ does the calculation automatically for us.

Q10

The regression shows a negative, statistically significant relationship between the distance to wildfires from the BGs and pro-environment policies. Each 100km (or 1km) increase in distance corresponds, on average, to a 0.04 (or 0.0004) percentage-point **drop** in the environmental index. Although the result is statistically significant at the 1% level, its magnitude is very low (less than 1% of the average index score). This suggests that California BGs have already formed a pro-environment consciousness: they still support environmental policies, even if they are not directly affected by wildfires.

Q11.

The covariance, because it shows how the two variables of interest change together. This change can be either positive or negative, indicating the sign of the slope, since variance can only take non-negative values.

Q12.

The intercept, approximately 0.65, suggests that BGs with wildfires at their centers have, on average, app. 65% of pro-environment support.

4 Policy Recommendation

To the Office of the Governor of California

From: Lucas I. Klotz, policy analyst

Subject: Wildfires public messaging

Wildfires have become a pressing issue in California due to climate change. The current administration has the opportunity to raise public awareness and enact policies to mitigate this problem. Our findings show that such environmental consciousness has already taken place across the state. Still, I offer one cost-effective action to further increase support.

Our analysis provides a key insight: support for pro-environment policies decreases as wildfires occur farther from a given Census block group (BG). In particular, our regression analysis, focusing mainly on Proposition 23 (2010), shows that each 100km increase in distance corresponds, on average, to a 0.04 percentage-point drop in the environmental index measuring the support for that Proposition. The result is statistically significant at the 1% level.

One can agree that, however, the magnitude is very low: less than 1% of the average index score, which is 0.63 in California. In other words, the share of the BG supporting a particular set of environmental policies is, on average, around 63%. What does this tell us?

This finding suggests that, on average, the state of California has already developed a pro-environment consciousness: the majority of residents in a particular BG still support environmental policies, even if they are not directly affected by wildfires.

Based on this finding, the Governor should have sufficient support to pass new environmental bills without additional concessions. If the administration wants to further increase support, I recommend a low-cost information campaign targeted at BGs less exposed to wildfires. Such ads would not hurt the state budget and could raise environmental awareness in areas that already lean pro-environment but still offer room for marginal gains, helping close the support gap with the BGs most affected by the fires.